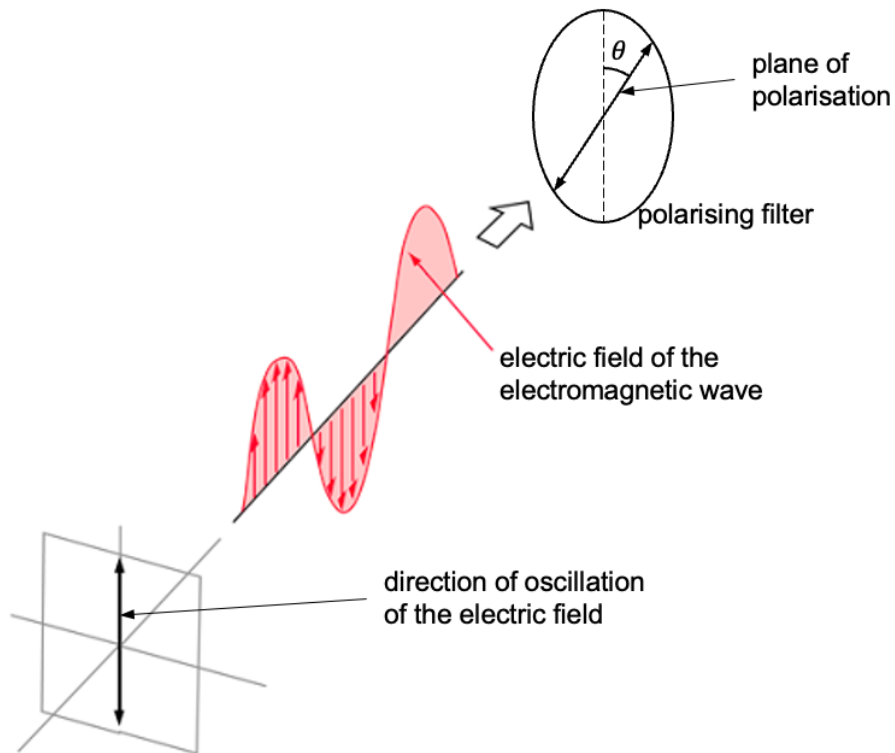


- 16 A plane-polarised electromagnetic wave is incident on a polarising filter as shown below.



The electric field of the wave before reaching the filter is vertical and the magnitude E of the field is

$$E = E_0 \sin \sin (\omega t) .$$

where E_0 is the amplitude and t is the time.

The plane of polarisation of the filter is at θ from the vertical.

What is the amplitude of the electric field emerging from the filter?

- A $E_0^2 \cos^2 \theta$
- B $E_0 \cos \theta$
- C $E_0 \sin \theta$
- D $E_0^2 \sin^2 \theta$